
x
[B,n,H]
Wqkv
Wg
Q, K, V = W·x
+ shortconv
gates βt, λ̄t
= Wg·x
shift
φ(qt)
xt = φ(kt−1)
(RAW shift)
B replays
window
N̄t(B) =
(1/Bt)Σxjvj^T
energy B
energy B
Ēt(B) =
(1/Bt)Σ‖xj‖²
ηt = β̄t/(Ēt(B)
+λt+ε)
window write
St = γtSt−1 +
ηt·N̄t(B)
γt = 1 −
min(ηtλt, 1−ε)
read
next
carry
ot = St^T·φ(qt)
(read-after-write)
St−1
[dx,dv]
Wo
O = Wo·ot
[B,n,H]